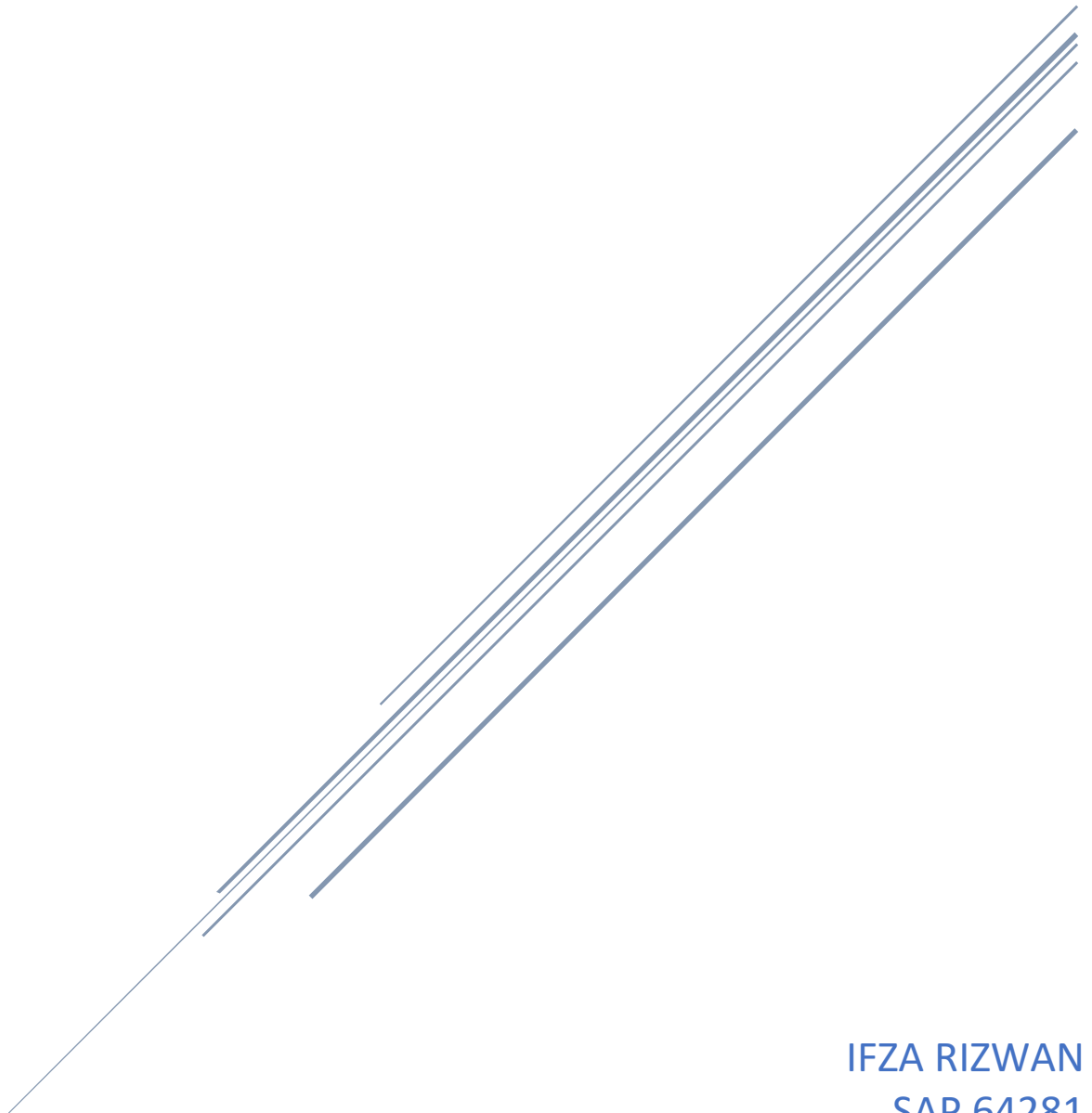


COMPUTER NETWORKS

LAB 11

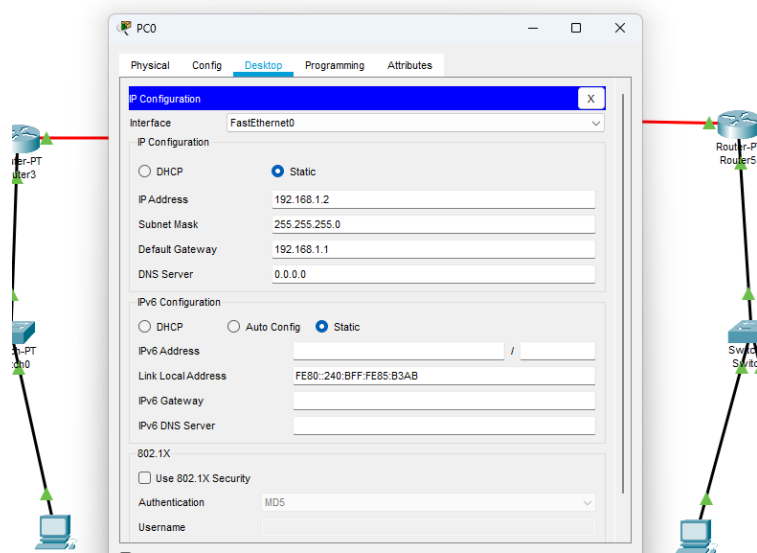


IFZA RIZWAN
SAP 64281

Task 1:

Perform all the tasks in the manual and take screenshot. (10 marks)

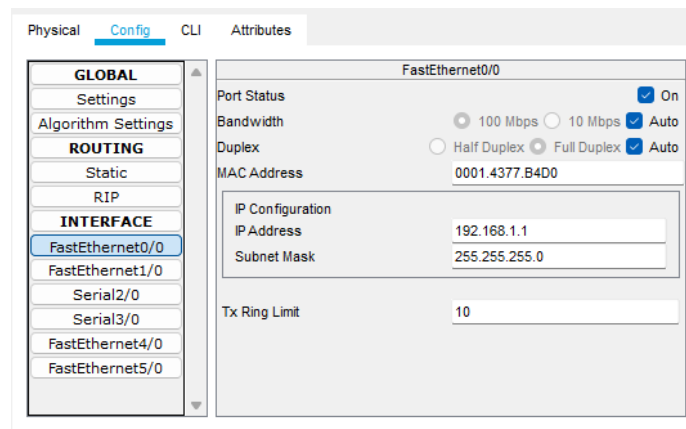
1. GIVE IP TO ALL PCs



Same method give ip and default gate way to all pcs the ips and default gateways ips are given in manual.

2. Now configure routers:

Router 0:



Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IP Address 10.0.0.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Router 1:

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth 100 Mbps 10 Mbps Auto

Duplex ☐ Half Duplex ☒ Full Duplex Auto

MAC Address 0002.174A.C388

IP Configuration

IP Address 192.168.2.1

Subnet Mask 255.255.255.0

Tx Ring Limit 10

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial2/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IP Address 10.0.0.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

Serial3/0

Port Status ☒ On

Duplex ☐ Full Duplex

Clock Rate 64000

IP Configuration

IP Address 11.0.0.1

Subnet Mask 255.255.255.252

Tx Ring Limit 10

Equivalent IOS Commands

Router 2:

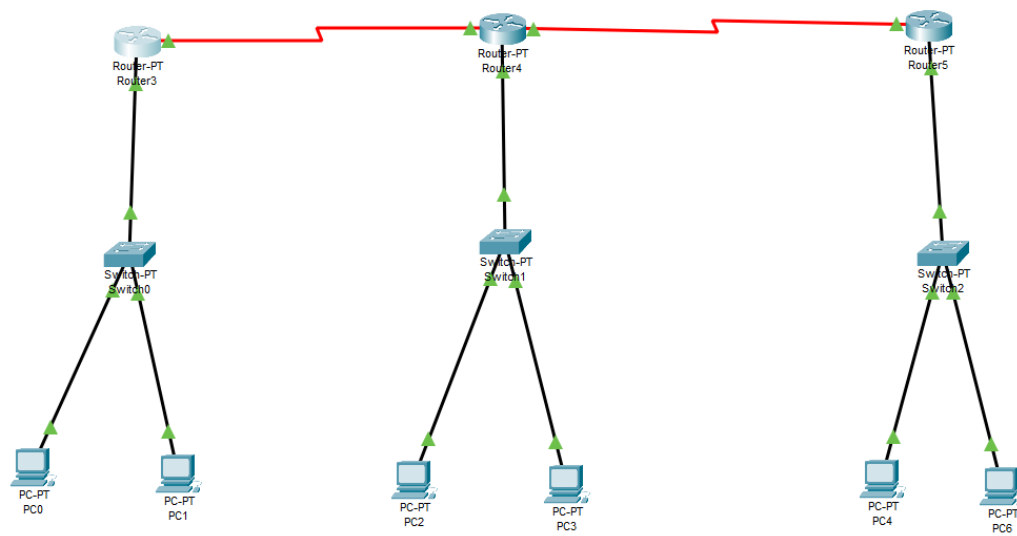
Physical **Config** CLI Attributes

GLOBAL	FastEthernet0/0
Settings	Port Status ☒ On
Algorithm Settings	Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto
ROUTING	Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto
Static	MAC Address 0060.2F19.C99A
RIP	IP Configuration
INTERFACE	IP Address 192.168.2.1
FastEthernet0/0	Subnet Mask 255.255.255.0
FastEthernet1/0	Tx Ring Limit 10
Serial2/0	
Serial3/0	
FastEthernet4/0	
FastEthernet5/0	

Equivalent IOS Commands

Physical **Config** CLI Attributes

GLOBAL	Serial2/0
Settings	Port Status ☒ On
Algorithm Settings	Duplex ☒ Full Duplex
ROUTING	Clock Rate 1200
Static	IP Configuration
RIP	IP Address 11.0.0.2
INTERFACE	Subnet Mask 255.255.255.252
FastEthernet0/0	Tx Ring Limit 10
FastEthernet1/0	
Serial2/0	
Serial3/0	
FastEthernet4/0	
FastEthernet5/0	



Ping the pc3 to pc4 router 1 to 2:

```
C:\>ping 192.168.3.2

Pinging 192.168.3.2 with 32 bytes of data:

Reply from 192.168.3.2: bytes=32 time=6ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126
Reply from 192.168.3.2: bytes=32 time=2ms TTL=126
Reply from 192.168.3.2: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 2ms

C:\>|
```

Pc0 to pc3:

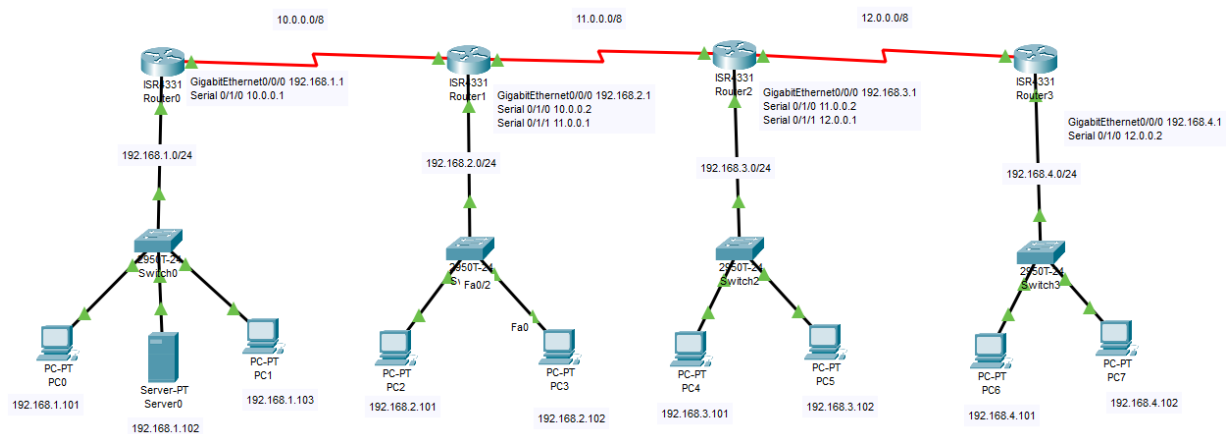
```
Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=6ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126
Reply from 192.168.2.3: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 2ms

C:\>|
```

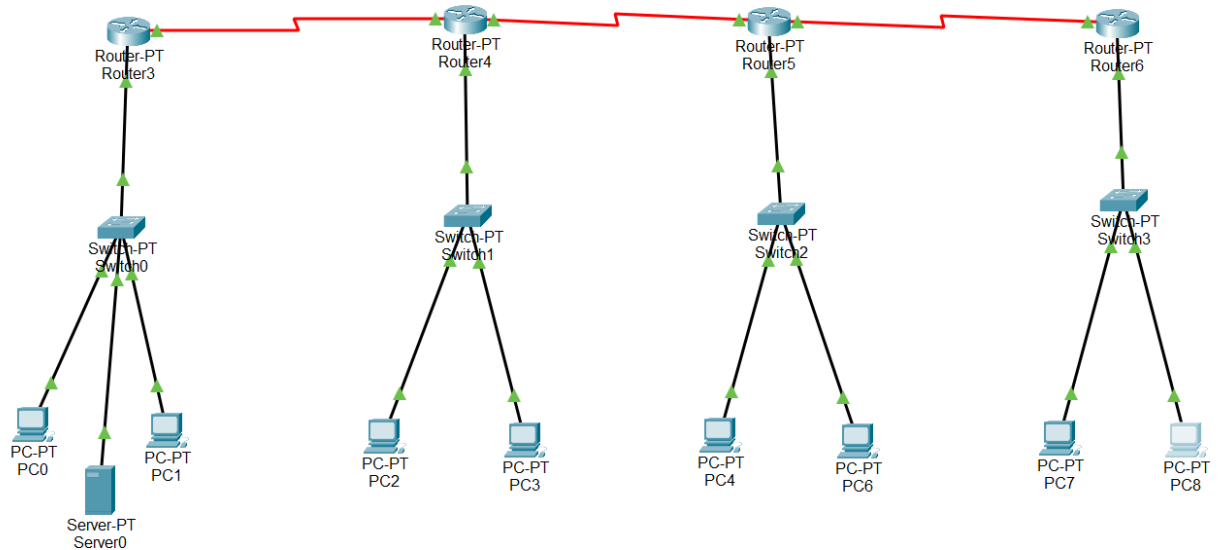
Task 2:



Create the above topology using rip. (10 marks)

HERE topology is created as the same as in above task all steps.

**ALL PCS AND ROUTERS ARE CONFIGURES SAME AS ABOVE
ADDITIONAL STEPS ARE STATED BELOW.**



ADDITIONAL STEP:

Server configuration especially ip assigning and subnet mask:

IP Configuration

IP Configuration

☐ DHCP
 ☒ Static

IPv4 Address
 192.168.1.4

Subnet Mask
 255.255.255.0

Default Gateway
 192.168.1.1

DNS Server
 0.0.0.0

IPv6 Configuration

☐ Automatic
 ☒ Static

IPv6 Address
 /

Link Local Address
 FE80::230:A3FF:FE46:6992

Default Gateway

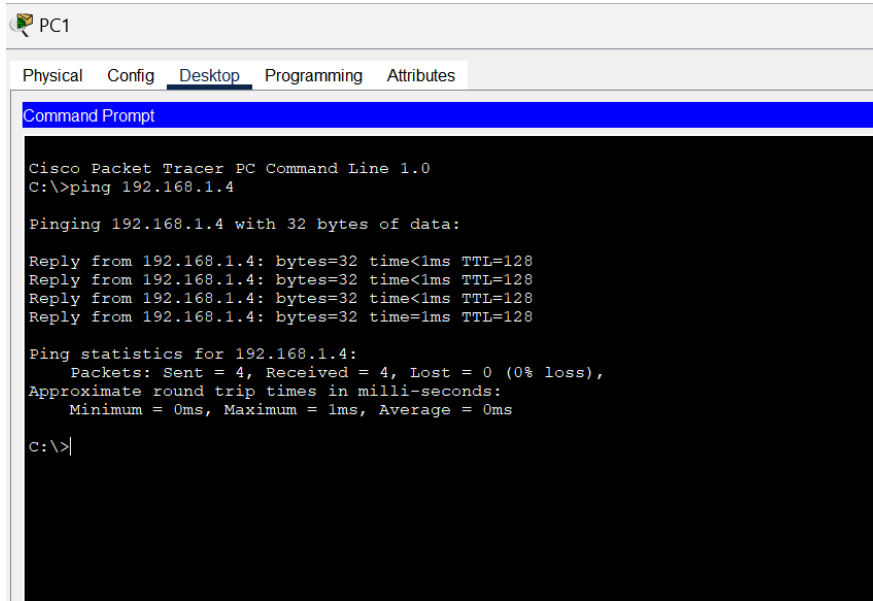
DNS Server

802.1X

☐ Use 802.1X Security

Authentication
 MD5

Now ping the server from inside the network:



The screenshot shows the Command Prompt window for PC1 in Cisco Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of a ping command to 192.168.1.4. The output shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 128. The ping statistics show 4 packets sent, 4 received, 0 lost (0% loss), and an average round trip time of 0ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

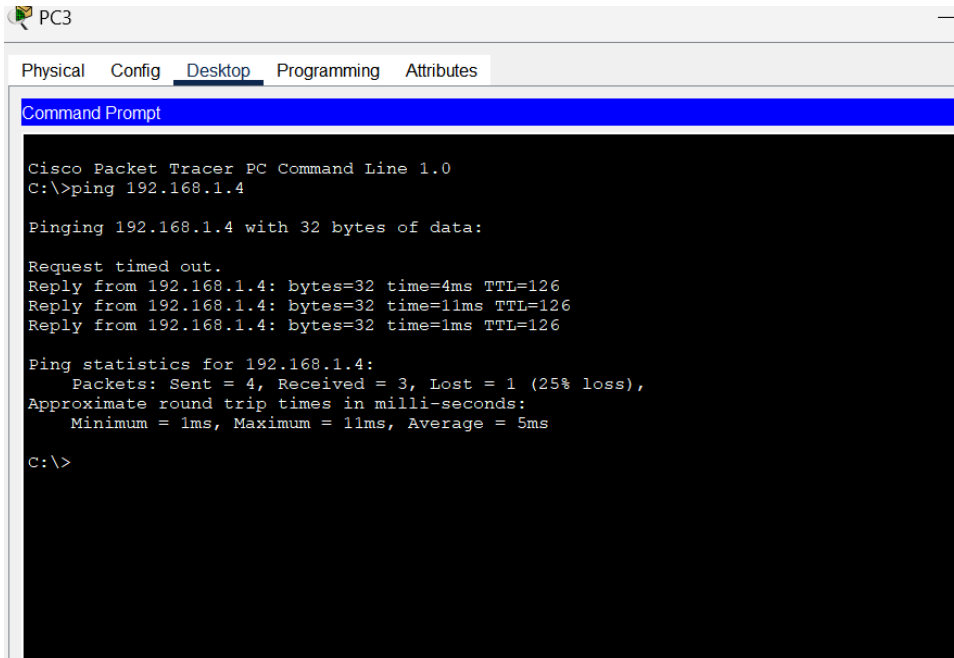
Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Now ping the server from outside the network:



The screenshot shows the Command Prompt window for PC3 in Cisco Packet Tracer. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt displays the output of a ping command to 192.168.1.4. The output shows a request timed out, followed by three successful replies with 32 bytes of data, a time of 4ms, 11ms, and 1ms, and a TTL of 126. The ping statistics show 4 packets sent, 3 received, 1 lost (25% loss), and an average round trip time of 5ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.4: bytes=32 time=4ms TTL=126
Reply from 192.168.1.4: bytes=32 time=11ms TTL=126
Reply from 192.168.1.4: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 5ms

C:\>
```

THE END